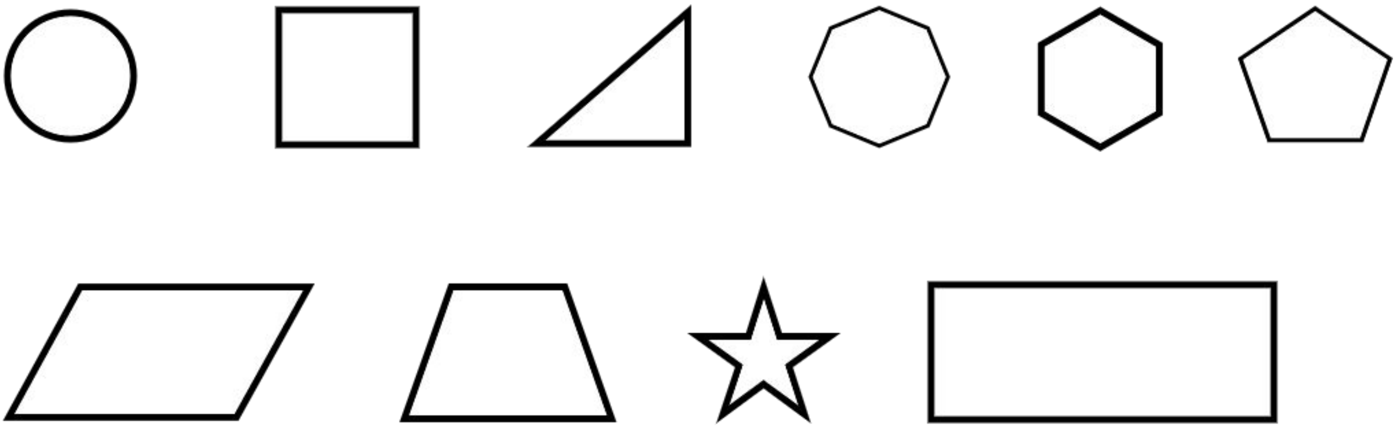
Checking understanding of perimeter and area

Task A: Defining a shape

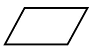
1) Pick one of the shapes and describe it to your partner.

How many clues did it take?

Which shape required the most clues?



There are many possibilities. Here are some examples:

- *My shape is not a **polygon**.* 
- *My shape has six sides.* 
- *My shape has four sides. There are two pairs of **parallel**, equal sides. The interior angles are not right angles.* 

Task B: Grouping shapes by properties

1) a) Put each **polygon** in the correct place.

Square

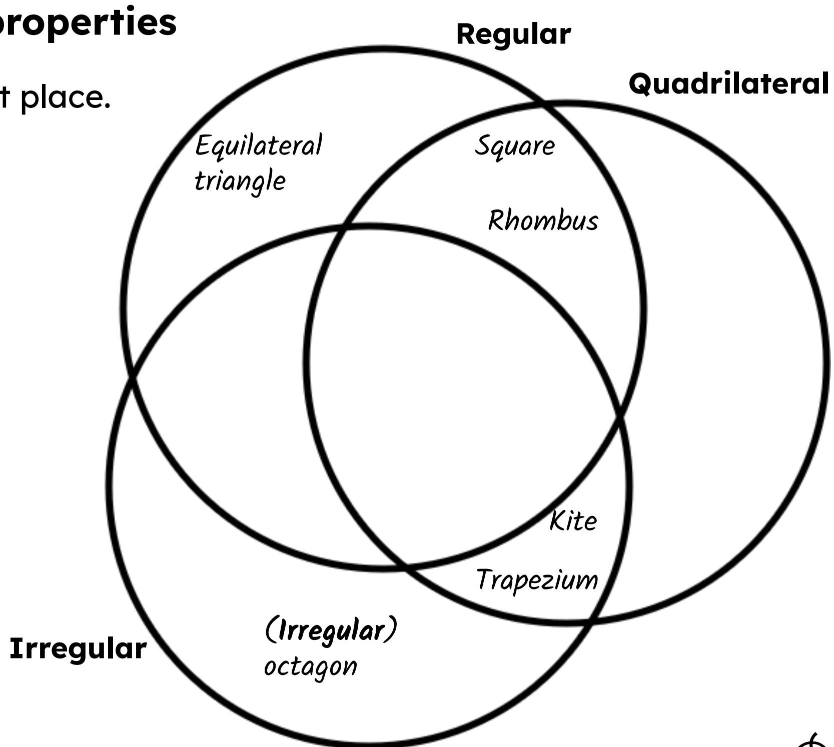
Trapezium

(**Irregular**) Octagon

Kite

Equilateral triangle

Rhombus



b) Why are some of the spaces empty?

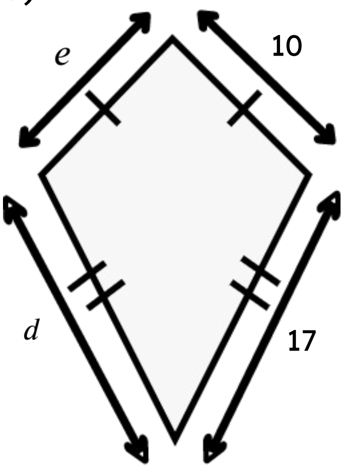
Some of the spaces are empty because a shape cannot be ***regular*** *and* ***irregular*** *at the same time.*

Quadrilaterals *must be either* ***regular*** *or* ***irregular*** *and so cannot be neither.*

Task C: Finding missing lengths with shape properties

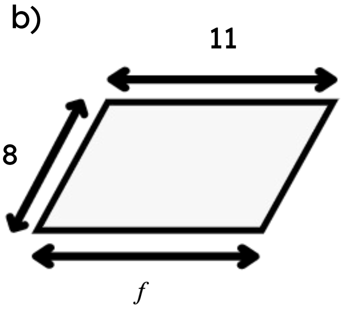
1) State the missing length for each shape.

a)



$d = 17$ $e = 10$

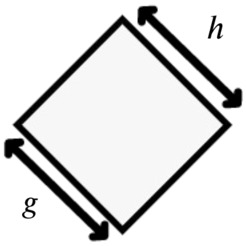
b)



This is a parallelogram.

$f = 11$

c)



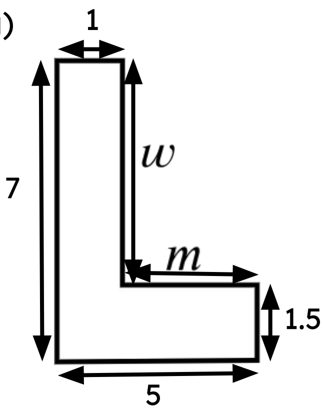
This is a 3 unit square.

$g = 3$ $h = 3$

Task D: Missing lengths in composite rectilinear shapes

1) Calculate the length of each marked side.

a)



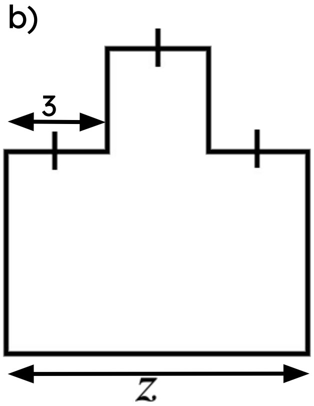
$m = 5 - 1$

$m = 4$

$w = 7 - 1.5$

$w = 5.5$

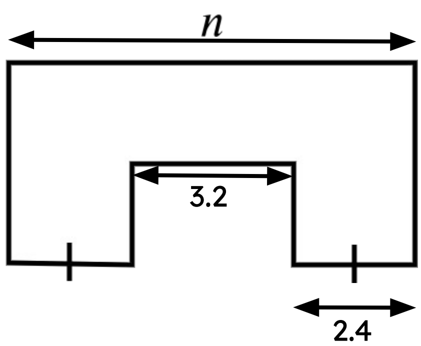
b)



$z = 3 + 3 + 3$

$z = 9$

c)



$n = 3.2 + 2 \times 2.4$

$n = 8$